UNIVERSITY OF TURKU

To Renounce Liberty? A Public-Goods Experiment on (Un)Democratic Reversal

Research Plan for Ethical Review

Hector Bahamonde[1,2]

[1]INVEST Research Flagship Centre, University of Turku, Finland
[2]Social Science Research Institute, Åbo Akademi University, Finland

Principal Investigator and project contact: Janne Tukiainen, Professor, Turku School of Economics, University of Turku; janne.tukiainen@utu.fi

Version 2.6 — proposed amendment — 20 October 2026

Abstract

Democratic backsliding often proceeds through lawful democratic choices that weaken citizens' control over what happens next. This project asks whether citizens vote to restore an approval right they supported surrendering once its crisis-based justification recedes. In a one-session public-goods experiment with 300 participants in Finland, five-person groups decide whether an unknown leader's proposal requires citizen approval or takes effect directly. Acting without approval initially combines a higher fund rate with assured implementation of the leader's equal allocation. Structural recovery improves public-service conditions and raises the approval-constrained rate to match direct implementation, while the leader's powers and decision rules remain unchanged. Comparing randomly assigned recovery and persistence pools identifies whether recovery causes previous supporters of direct implementation to vote for group approval. The experiment captures a demand-side microfoundation of executive aggrandizement: citizens can use one democratic decision to weaken a later constraint and another to vote to take that control back.

Experiment prototype (production phase):
`https://panel-lab-experiment.onrender.com`

1 Research Question and Contribution

Democratic backsliding depends not only on leaders who weaken constraints but also on citizens who authorize them. This project studies a narrow demand-side microfoundation of executive aggrandizement: the citizen-level authorization through which a lawful democratic decision weakens a constraint on one-person authority. It asks: *after a crisis recedes, do citizens who supported decisions without their approval vote to restore that right?* This within-person change in institutional support is the **(*un*)democratic reversal** the project seeks to explain.

We build on the democratic backsliding and public-good games literatures. Contemporary backsliding often advances without ending elections: citizens may support elected officials or lawful rule changes that reduce the approval, oversight, or veto constraints governing later decisions (Bermeo 2016). The politically relevant step is therefore not necessarily an explicit vote against democracy, but a *democratic* vote that makes subsequent authority *less* constrained. Threat and policy benefits can make citizens more tolerant of such constraints being weakened (Hetherington and Suhay 2011; Goodman 2021; Saikkonen and Christensen 2023), while laboratory evidence shows that gridlock can increase support for executive special powers (Forteza, Mussio, and Pereyra 2024).

This literature mainly explains movement toward less constrained authority. Our contribution is to follow the reverse movement: once the crisis-based benefit recedes, do the same citizens vote to restore the approval right they previously supported surrendering? Observing that second choice turns reversal into a consequential decision over an institutional constraint rather than a stated evaluation of democracy.

Public-good games supply the economic reason for weakening approval. Voluntary provision is vulnerable to free riding—keeping one's points while others fund the shared service—and declining cooperation, whereas centralized enforcement can coordinate contributions (Fehr and Gächter 2000; Fischbacher and Gächter 2010; Hamman, Weber, and Woon 2011). Experiments also show that voting over institutions changes their performance (Dal Bó, Foster, and Putterman 2010; Sutter, Haigner, and Kocher 2010; Markussen, Putterman, and Tyran 2014). Our experimental design joins these literatures by making citizen approval both institutionally meaningful and economically costly.

In each round, five citizens begin with 20 points each and vote between two rules for managing a shared public-services fund. Under both rules, the software selects one of the five citizens uniformly at random only after the vote, so each has the same one-in-five chance of leading that round. The leader proposes how many points every citizen places in the fund and whether any collected points move to the leader's payoff. Fund points are multiplied and the resulting group return is divided equally.

Under the approval-constrained rule (D), at least three of the other four citizens must approve the complete proposal. Rejection returns the group to voluntary individual allocations. Under automatic implementation (A), the same proposal takes effect without that approval vote. A matching pool contains ~~10 or 15~~ 5, 10, or 15 citizens~~ and is reshuffled into two or three temporary groups of five before each round~~. Larger pools are reshuffled into two or three groups of five before each round, while a five-person pool plays together throughout. Thus, the democratic ballot selects an approval rule, not a known leader or proposal.

Everyone begins with a public-service crisis: "*Parliament has failed to pass a budget for eight months, healthcare waiting times have doubled, and planning for public services has stalled.*" During Block 1, each fund point creates 1.50 group points under D and 2.50 under A. Direct implementation therefore has a one-point productivity premium and guarantees that the leader's proposal is implemented. Afterward, each matching pool receives **institutional recovery** or **institutional persistence**. Recovery is one structural package: public-service conditions improve and the D rate rises to 2.50. Persistence retains the crisis and the 1.50 rate. The A rate and decision rule remain fixed.

The payoffs make the choice consequential; **the *allocation of decision rights* connects it to democratic backsliding**. A citizen who votes for A supports surrendering an approval vote before knowing the leader or proposal. A later vote for D supports restoring that safeguard after the crisis-based productivity advantage disappears. *The experiment therefore observes whether citizens vote to take back a specific constraint on authority, rather than inferring reversal from a survey declaration.*

In simple terms

Five citizens vote before seeing who will lead or what the leader will propose. Under D, three of the other four citizens must approve the proposal; rejection returns everyone to individual fund allocations. Under A, the proposal takes effect directly. In Block 1, these methods use rates of 1.50 and 2.50, respectively.
After Block 1, recovery improves public-service conditions and raises D to 2.50; persistence leaves the crisis and rate gap in place. Among previous A supporters, a later D vote supports restoring the approval right. Comparing recovery with persistence identifies how much the structural recovery package changes that reversal.

Central insight

A decision can be democratically authorized while weakening the democratic constraints that govern what happens next. This resembles contemporary backsliding, which often proceeds through elections and formally lawful institutional changes rather than an outright abolition of democracy (Bermeo 2016).
In the experiment, the first majority vote is democratic. A citizen who chooses A votes to let an unknown leader's later allocation and transfer take effect without the other citizens' approval. ~~That vote~~ A majority for A removes a specific constraint while leaving the institutional ballot itself intact. A later vote for D supports restoring the approval right the citizen previously voted to surrender.

What matters is that citizens use one democratic decision to weaken their control over subsequent decisions—and can later vote to take that control back.

2 Theory

2.1 Crisis and the productivity premium

A crisis can make action without approval attractive when ordinary procedures produce delay or failed coordination. The experiment gives this argument material consequences. In Block 1, proposals requiring approval (D) use a 1.50 fund rate, while proposals implemented directly (A) use 2.50. Under D, rejection also returns the group to voluntary contributions and possible free riding. Direct implementation therefore offers both a productivity premium and assured implementation of the leader's equal allocation (Fehr and Gächter 2000; Fischbacher and Gächter 2010; Hamman, Weber, and Woon 2011).

The personal-transfer option makes the institutional constraint consequential. The leader may propose moving up to 20 fund points to their own payoff under either procedure. Under D, the other citizens see that transfer before implementation and can reject it; under A, they observe it only after it has taken effect. The experiment therefore concerns approval over discretion, not the prevalence of corruption in Finland.

H1: Authorization under crisis. During Block 1, ~~citizens will increasingly support~~ better outcomes under direct implementation (A) ~~when its productivity and coordination advantages~~

~~produce better outcomes;~~ are associated with greater subsequent support, while larger observed transfers ~~will reduce subsequent support~~ are associated with lower support. These are descriptive expectations about experience within the crisis environment.

2.2 Citizen authorization and democratic constraint

The institutional ballot occurs before leader selection and before the proposal. A vote for A therefore ~~authorizes~~ supports a rule, not a known person or action~~: the citizen relinquishes~~. The ballot supports relinquishing participation in approving the next binding decision. A vote for D ~~retains~~ supports retaining that approval right. **Thus, our design provides an abstract microfoundation of *executive aggrandizement*, in which a *democratic* choice *weakens* the constraint governing what follows** (Bermeo 2016; Forteza, Mussio, and Pereyra 2024).

H2: Approval and discretion. Approval will decline as proposed personal transfers increase. ~~Anticipating rejection, leaders operating under D will propose smaller transfers than leaders whose proposals take effect directly. Because groups choose the institutional rule themselves, these patterns describe behavior within the chosen rules rather than a separate randomized effect.~~ When leaders expect citizens to reject transfers despite the earnings from an approved proposal, leaders under D will propose smaller transfers than those under direct implementation. These are behavioral expectations within endogenously chosen rules, not a separate randomized effect.

2.3 Recovery and democratic reversal

Recovery is assigned at the matching-pool level. Pool g contains ~~10 or 15~~ 5, 10, or 15 participants~~, who are reshuffled into two or three temporary groups of five before each round~~. Larger pools are reshuffled into two or three temporary groups of five before each round, while a five-person pool remains together; within each group, three votes produce a majority. The entire pool receives the same Block-2 condition: $T_g = 1$ denotes recovery and $T_g = 0$ persistence. Let m_{gb} denote the approval-constrained fund multiplier in pool g and block b. The productivity premium for direct implementation is

$$\delta_{gb} = 2.50 - m_{gb} \tag{1}$$

Everyone faces $\delta_{g1} = 2.50 - 1.50 = 1$ in Block 1. Recovery is deliberately implemented as one structural package: participants learn that public-service conditions improved and m_{g2} rises to 2.50, setting $\delta_{g2} = 0$. Persistence reports that the crisis continues and keeps $m_{g2} = 1.50$, so $\delta_{g2} = 1$. The leader-selection rule, proposal set, approval threshold, rejection-to-voluntary-allocation rule, and A rate remain fixed.

The recovery–persistence comparison identifies the total causal effect of structural recovery on (un)democratic reversal. Among final-round Block-1 supporters of A, the analysis compares later D votes in recovery pools ($T_g = 1, \delta_{g2} = 0$) with those in persistence pools ($T_g = 0, \delta_{g2} = 1$). The public-service description and production rate jointly define recovery: the text states what changed, while the rate makes that change materially consequential inside the game. The estimand therefore concerns their combined effect rather than separate framing and payoff effects. Persistence supplies the counterfactual in which both the crisis and premium remain.

H3: Immediate restoration. Among participants who vote for A in Block 1's final round, recovery will increase the probability of voting for D in Block 2's first round.

H4: ~~Sustained reversal~~ Short-run reversal. Among the same participants, recovery will increase the probability of voting for D in Block 2's final round (Round 5).

3 Experimental Design

3.1 Overview

Each laboratory cohort completes one session lasting approximately 60 minutes, with ~~two ten-round blocks~~ ten paid rounds in Block 1 and five in Block 2. ~~Subject to ethical review in September 2026~~ Following the favorable ethical-review statement of 30 September 2026 (case TY/1722/2026), and subject to any required amendment review, data collection is planned for 1–31 October 2026 at the UTU Choice Lab decision-making laboratory~~ and may also be conducted at Tampere University's Decision-making Laboratory (DMLab) to accelerate recruitment~~. UTU Choice Lab has 24 regular seats plus one accessible seat~~, while DMLab has 20~~; production rosters are capped at 20. Any later extension of data collection to another site will be submitted for ethical review as an amendment before collection begins there. Participants remain in matching pools of ~~10 or 15~~ 5, 10, or 15~~ and are reshuffled into anonymous five-person groups before every paid round~~. Larger pools are reshuffled into anonymous five-person groups before every paid round; a five-person pool plays together throughout. After Block 1, each entire pool is assigned to recovery or persistence, learns whether public-service conditions improved or remained strained, and sees its Block-2 fund rates. Participants therefore interact repeatedly~~ without forming stable five-person teams~~, with rematching whenever the roster permits. In the figures, p_i identifies participant i.

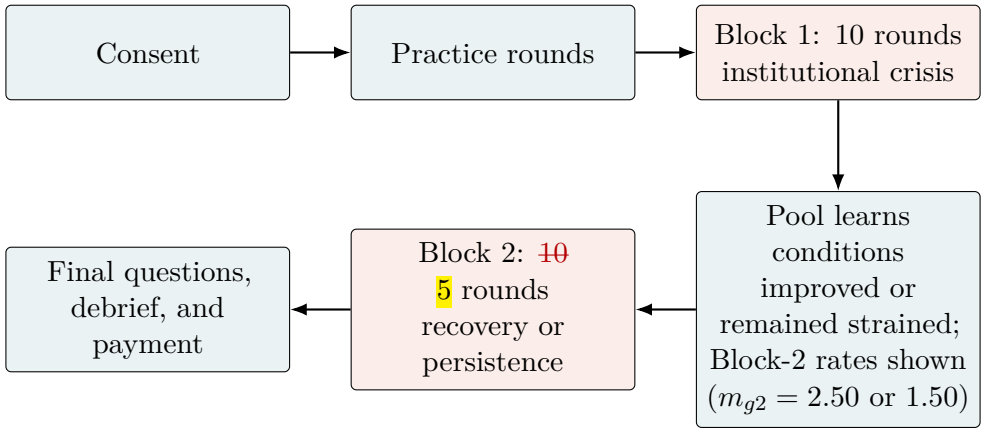


Figure 1: One-session sequence. Each paid round includes an institutional ballot, leader proposal, implementation, feedback, and rematching in larger pools.

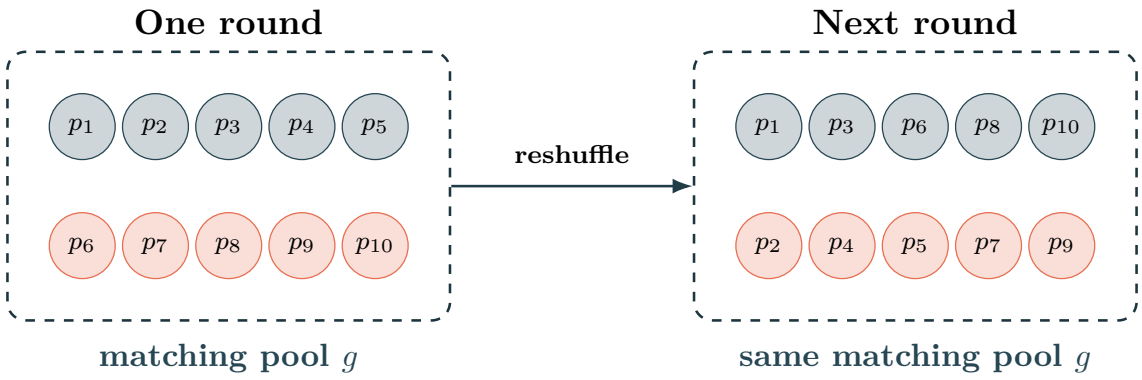


Figure 2: Anonymous rematching within a matching pool (ten-person example).
Notes: Each participant p_i remains in pool g. A ten-person pool forms two temporary groups per round; a fifteen-person pool forms three. A five-person pool retains the same playing group. Recovery or persistence applies to the whole pool in Block 2.

In simple terms

~~Matching pools permit interaction without fixing the same five-person team~~ Matching pools contain interaction within one treatment environment. ~~Round-by-round reshuffling limits stable reputations, collusion, and retaliation, while keeping participants inside one treatment environment~~ Larger pools are reshuffled each round, limiting stable-team histories. ~~Ten-person pools are used whenever the completed cohort can be divided that way; one fifteen-person pool accommodates a remaining set of five while retaining rematching~~ Ten participants form one ten-person pool, fifteen form one fifteen-person pool, and twenty form two ten-person pools. A five-person session remains feasible, with the same members playing throughout.

Reversal is measured for individuals, but treatment (T_g) is assigned to pools. Statistical uncertainty is therefore clustered by pool~~; temporary groups organize play but are not treatment units~~. Playing groups are not additional treatment units. ~~Ten-person pools remain the default because they yield more independent treatment assignments for a fixed sample~~ The analysis accounts for pool size and, where estimable, checks results excluding five-person pools, where shared histories may be stronger.

Table 1: Session formation from the completed laboratory roster

People who arrive	Matching pools	Five-person groups per round	Participate	Released
20	$10 + 10$	$2 + 2 = 4$	20	0
16–19	15	3	15	1–4
15	15	3	15	0
11–14	10	2	10	1–4
10	10	2	10	0
6–9	5	1	5	1–4
5	5	1	5	0
Fewer than ~~10~~ 5	—	—	0	All

Notes: ~~A matching pool is not a playing group~~ A matching pool contains one or more playing groups. Each ten-person pool forms two temporary five-person groups per round; a fifteen-person pool forms three. At 20 participants, two separate ten-person pools each form two groups. ~~Groups dissolve and are reshuffled after every round~~ Larger pools are reshuffled every round, while a five-person pool plays together throughout; participants never enter or change pools after paid play begins.

3.2 The institutional-choice public-good game

Each citizen starts with 20 points and privately votes between an approval-constrained rule (D) and automatic implementation (A). Let $V_{i,b,r} \in \{\mathsf{D}, \mathsf{A}\}$ denote citizen i's ballot in block b and round r. Three votes select the binding rule. The software then selects each group's leader by an independent uniform draw among its five citizens. Selection occurs under both rules and only after the institutional ballot, giving every citizen the same ex ante chance in every round without making past service predict the next draw.

Table 2: Implemented design parameters

Feature	Implemented value
Laboratory session	One synchronized visit, approximately 60 minutes
Paid rounds	~~10 per block (20 total)~~ 10 in Block 1 and 5 in Block 2 (15 total), preceded by unpaid practice
Group and matching pool	5-person groups; stable ~~10- or 15-person~~ 5-, 10-, or 15-person matching pools
Rematching	Anonymous rematching within ~~pool~~ larger pools before every paid round; fixed membership in a five-person pool
Round endowment	20 points per participant
Approval-constrained multiplier	1.50 in Block 1; 2.50 recovery / 1.50 persistence in Block 2
Automatic-implementation multiplier	2.50 in both blocks and conditions
Approval	At least 3 of the 4 nonleaders; a 2–2 split rejects the proposal
Rejection	Voluntary individual contributions at the approval-constrained rate
Personal transfer	0–20 fund points, limited by the points collected
Performance bonus	Points earned in one randomly selected round from each block (one of 10 and one of 5)
Maximum performance bonus	EUR 5.00
Show-up payment	EUR 10.00
Maximum total payment	EUR 15.00

Selected-round points are converted linearly into money at EUR 5 per 120 points. Because a selected round can yield at most 60 points, the two selected rounds produce a bonus of at most EUR 5. Each Block-1 round has a one-in-ten payment chance; each Block-2 round has a one-in-five chance. Selecting one round from each block keeps every decision consequential without paying all rounds or encouraging participants to offset one decision with another.

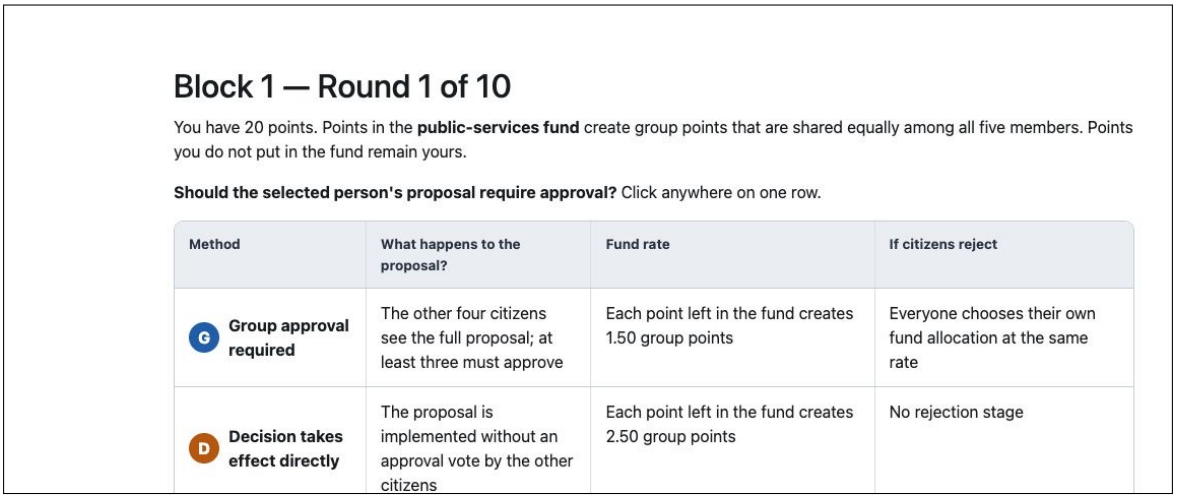


Figure 3: Participant screen for choosing whether the leader's proposal requires approval.

Leader proposal. Under either rule, the leader proposes an equal allocation $t \in \{0, \ldots, 20\}$ and a personal transfer $\rho \in \{0, \ldots, 20\}$, with $\rho \leq 5t$. Let π denote a citizen's payoff in that

round. Under A, the proposal takes effect directly, and a nonleader earns

$$\pi^{A}_{igbr} = 20 - t + \frac{2.50}{5}(5t - \rho), \tag{2}$$

and the leader earns $\pi^{A,L}_{igbr} = \pi^{A}_{igbr} + \rho$.

Approval-constrained procedure (D). The four nonleaders observe (t, ρ) and vote privately; let $q_{igbr} = 1$ denote approval. The proposal passes when at least three approve. If it passes, the payoffs above apply with m_{gb} replacing 2.50. If it fails, each citizen chooses $c_{igbr} \in \{0, \ldots, 20\}$ and earns

$$\pi^{D}_{igbr} = 20 - c_{igbr} + \frac{m_{gb}}{5} \sum_{j=1}^{5} c_{jgbr}. \tag{3}$$

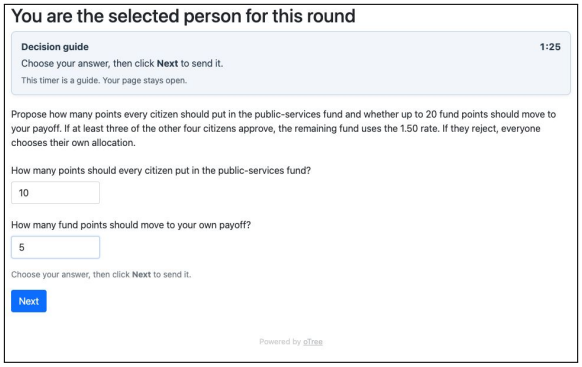
The rejected transfer is not implemented. Everyone then sees the proposal, approval total, implemented allocation, transfer, and own payoff.

The monetary benchmark assumes participants maximise their own earnings and approve proposals that beat the 20-point free-riding fallback. A leader then chooses an allocation of 20 and a transfer of 20. An approved crisis proposal pays 44 to the leader and 24 to each nonleader; direct implementation pays 60 and 40. After recovery, approval pays the same 60 and 40. Thus, the transfer cap need not induce rejection under this benchmark, while recovery removes the earnings difference between the approved rules. Observed adaptation over five follow-up rounds is an empirical question rather than a predicted convergence speed.

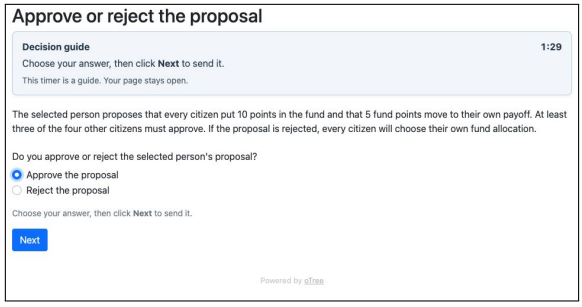
In simple terms

Rejection restores the standard **public-goods dilemma**. If one citizen contributes 1 point and four contribute 0, the contributor earns $19 + \frac{1.50}{5} = 19.30$ during the crisis, while each free-rider earns 20.30. Under recovery, these payoffs become 19.50 and 20.50. Contributing helps the group but costs the contributor; direct implementation avoids that risk, while approval lets citizens block an unwanted proposal.

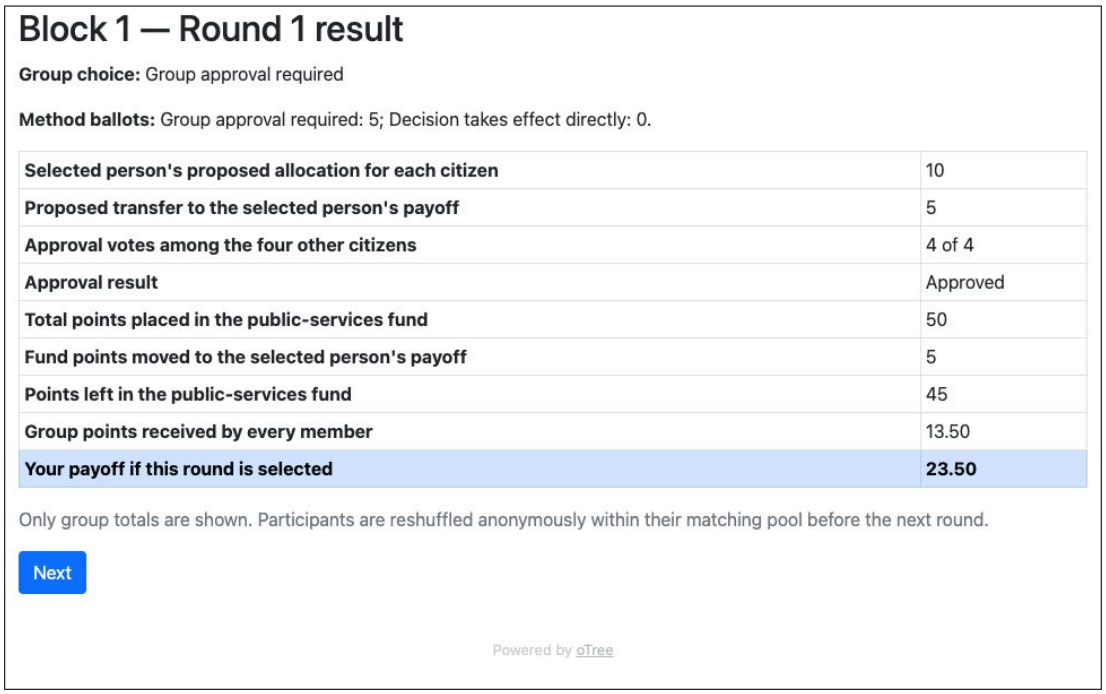
(a) Leader proposal



(b) Citizen approval



(c) Proposal, approval, and payoff feedback



Block 1 — Round 1 result

Group choice: Group approval required

Method ballots: Group approval required: 5; Decision takes effect directly: 0.

Selected person's proposed allocation for each citizen	10
Proposed transfer to the selected person's payoff	5
Approval votes among the four other citizens	4 of 4
Approval result	Approved
Total points placed in the public-services fund	50
Fund points moved to the selected person's payoff	5
Points left in the public-services fund	45
Group points received by every member	13.50
Your payoff if this round is selected	**23.50**

Only group totals are shown. Participants are reshuffled anonymously within their matching pool before the next round.

Next

Powered by oTree

Figure 4: Proposal, approval, and round feedback. Entered amounts are illustrative.

How incentives map onto reversal. During Block 1, A combines a 2.50 rate with guaranteed implementation; D uses 1.50 and may return to voluntary contributions after rejection. Recovery jointly reports improved public-service conditions and raises D to 2.50, while persistence retains the crisis and rate gap. The proposal set, approval rule, rejection procedure, and A rule remain unchanged. The recovery–persistence contrast therefore estimates the total effect of this structural package.

From institutional choice to (un)democratic reversal. A citizen who moves from A to D first votes to let an unknown leader's proposal take effect without approval and later votes to restore that approval. The public-good incentives make this act consequential: A still guarantees that the leader's equal allocation takes effect after recovery, while D may produce rejection and free riding. A higher switching rate under recovery shows that structural change can cause citizens to withdraw support for a weakly constrained decision rule. Importantly: *reversal is observed as a vote to restore a decision right, not as a declaration of democratic sentiment.*

A simple payoff comparison

Suppose the leader proposes $t = 10$ and $\rho = 0$, and the proposal is approved under D. Holding that decision fixed gives:

	Approval required (D)	Direct implementation (A)
Crisis or persistence	25 points each	35 points each
Recovery	35 points each	35 points each

Recovery makes an approved D proposal equally productive; it does not make it pay more. A still guarantees implementation. **The question is: *when approval becomes equally productive, do citizens vote to restore it?***

Participants report expected payoffs and transfers before the final Block-1 ballot and the first and final Block-2 ballots. Expected payoffs assess whether they understood the rules' relative performance; expected transfers trace beliefs about how leaders will use their discretion.

In simple terms

Five citizens cast private institutional ballots ($V_{i,b,r}$), after which the software selects a leader. That leader proposes (t, ρ). Under D, three of four nonleaders must approve; rejection activates individual contributions (c_{igbr}). Under A, the proposal takes effect directly. Everyone sees the result and payoff before ~~rematching within pool g~~ the next round. Larger pools are then reshuffled within g.

3.3 Block 1: crisis and leader authorization

After consent, participants practice proposing, approving, rejecting, and making individual allocations after rejection, then answer five comprehension questions. The interface never calls either method democratic or undemocratic.

Block 1 begins with the public-service crisis described above. Each fund point produces 1.50 under D and 2.50 under A. The leader may propose the same allocation and transfer under either rule; what differs is whether the other citizens can reject it.

Each paid round repeats the same sequence: institutional ballot, leader proposal, approval or direct implementation, any individual allocation following rejection, feedback, and rematching in larger pools. Expectations are elicited before the final ballot, and one Block-1 round is selected for payment.

3.4 Block 2: structural change and reversal

After Block 1, pools are assigned to *recovery* or *persistence*. Recovery jointly reports that the budget passed, waiting times are falling, and planning resumed, and raises the D multiplier from 1.50 to 2.50. Persistence retains the crisis and 1.50 rate. The A rate, leader selection, proposal set, approval threshold, and rejection procedure remain unchanged. Together, the updated description and rate form the structural-recovery treatment.

The first Block-2 ballot measures immediate support for restoring approval; the final ballot ~~measures sustained reversal~~ records support after four completed Block-2 rounds and is primary. Their joint pattern describes short-run persistence, delayed restoration, or abandonment of an immediate switch. Expectations are elicited before both ballots, and questions about democratic constraints appear only after paid decisions. One Block-2 round is selected for payment.

4 Outcomes, Process Measures, and Estimands

Let $\mathbb{1}\{\cdot\}$ equal 1 when the condition inside braces is true and 0 otherwise. The final Block-1 ballot defines endorsement of automatic implementation as

$$A_{i1} = \mathbb{1}\{V_{i,1,10} = \mathsf{A}\}. \tag{4}$$

For the theoretically relevant subgroup with $A_{i1} = 1$, reversal is

$$R_i = \mathbb{1}\{V_{i,2,\text{~~10~~}5} = \mathsf{D}\}. \tag{5}$$

The primary estimand is the average treatment effect of structural recovery among Block-1 supporters of automatic implementation:

$$\tau_R = \mathbb{E}[R_i(1) - R_i(0) \mid A_{i1} = 1], \tag{6}$$

where $R_i(1)$ and $R_i(0)$ are participant i's potential outcomes under recovery and persistence. Recovery includes both improved public-service conditions and the 2.50 D rate; τ_R is the total effect of that package. Because A_{i1} is measured before assignment, subgroup membership is fixed before treatment. The primary randomization test ~~repeatedly reassigns recovery according to the prespecified pool-level procedure to evaluate how unusual the observed contrast is~~ evaluates the sharp null of no recovery effect for any eligible citizen by reassigning treatment according to the prespecified pool-level procedure. Because τ_R describes the average eligible citizen, the primary estimate weights eligible participants equally; a secondary analysis gives each pool equal weight. Regression estimates cluster uncertainty by pool and include ~~a prespecified indicator for fifteen-person pools~~ prespecified pool-size indicators. Where estimable, a sensitivity analysis excludes five-person pools. Differences between fixed and rematched pools are descriptive because attendance, not randomization, determines pool size. Missing final ballots are reported by condition, with sensitivity checks retaining all randomized eligible citizens in the accounting.

Immediate reversal is

$$R_i^I = \mathbb{1}\{V_{i,2,1} = \mathsf{D}\} \quad \text{for } A_{i1} = 1. \tag{7}$$

The same comparison applied to R_i^I estimates immediate support for restoration. The final-round outcome R_i remains primary because it records ~~whether that choice persists~~ support after experience with the assigned environment. It includes both immediate switches maintained until Round 5 and switches made later.

In simple terms

The primary estimand (τ_R) asks: among citizens who voted for direct implementation in Block 1's final round ($V_{i,1,10} = \mathsf{A}$; $A_{i1} = 1$), how much more often does recovery ($T_g = 1$), rather than persistence ($T_g = 0$), lead them to vote for group approval in Block 2's final round ($V_{i,2,\text{~~10~~}5} = \mathsf{D}$; $R_i = 1$)? Only one potential outcome is observed per citizen, so random assignment identifies the difference. Assignment and inference operate at pool level.

For each leader i, t_{igbr} is the proposed allocation and ρ_{igbr} the proposed transfer. A transfer proposal is $M_{igbr} = \mathbb{1}\{\rho_{igbr} > 0\}$, with intensity $\rho_{igbr}/(5t_{igbr})$ when $t_{igbr} > 0$. Thus, $t = 10$ and $\rho = 5$ proposes moving 10% of the 50 collected points. Approval ballots q_{igbr}, proposal passage, contributions after rejection, and subsequent institutional ballots provide ~~descriptive~~ process evidence. ~~Because groups choose their rule, these measures do not identify separate causal effects of recovery~~ Comparisons between the endogenously chosen rules are descriptive and do not isolate the effect of requiring approval.

Immediate reversal R_i^I records the first response to the structural package, before Block-2 interaction (see Figure 5). The primary outcome R_i records Round ~~10~~ 5 after ~~nine~~ four completed

rounds. Secondary outcomes include late-round automatic-implementation support (Rounds 3–5) and realized group-level restoration. Persistence refers to choices within this session, not durability beyond it.

Expected payoffs assess whether participants understood how recovery changed the rules' relative performance, while expected transfers trace beliefs about leader discretion. After all paid choices, eleven process items check perceived crisis severity and recovery, expected institutional performance, and views on approval, equal allocation, and transfers. A separate optional twelve-item background module records age group, gender, education, household financial strain, party voted for in Finland's 2023 parliamentary election, left–right placement, whether democracy is preferable to other forms of government, satisfaction with democracy in Finland, evaluation of a strong leader unconstrained by parliament and elections, external political efficacy, generalized trust, and willingness to take risks. These prespecified measures capture partisan and ideological trade-offs, distinguish principled democratic support from evaluations of democratic performance, and benchmark cooperative and risk-related predispositions (Graham and Svolik 2020; Saikkonen and Christensen 2023; Claassen 2020; Dohmen et al. 2011; Glaeser et al. 2000). They are collected only after paid decisions and support ~~precision and~~ descriptive heterogeneity analyses; they do not alter the randomized estimand or independently identify a treatment mechanism. Primary precision adjustment uses only prespecified characteristics fixed before treatment and Block-1 measures, not potentially treatment-responsive attitudes collected afterward.

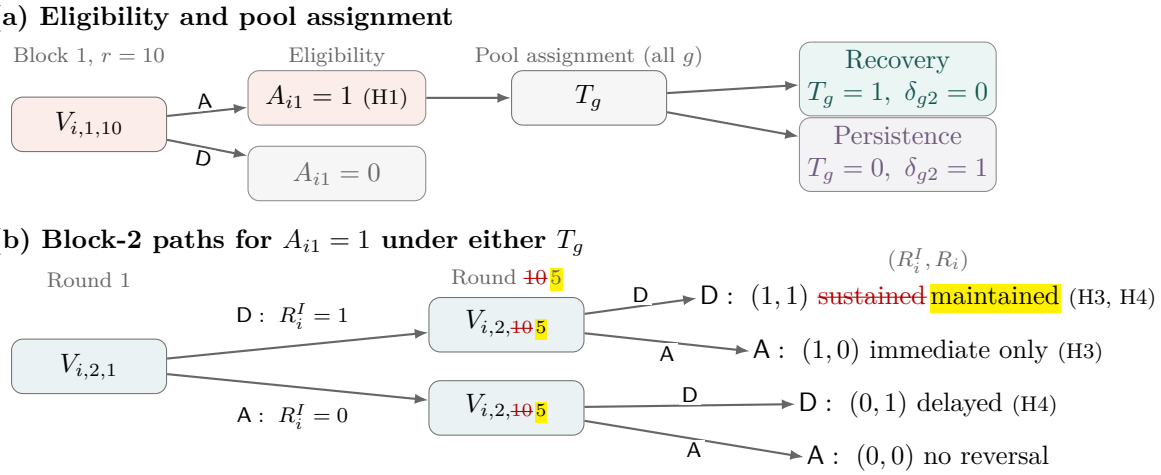


Figure 5: Decision tree for assignment and reversal. Each pool is assigned one value of T_g; the primary analysis follows participants with $A_{i1} = 1$.

Notes: The displayed ballots are $V_{i,1,10}$ (final Block 1), $V_{i,2,1}$ (first Block 2), and $V_{i,2,}$~~10~~5 (final Block 2). The ordered pair (R_i^I, R_i) records the two Block-2 outcomes, with 1 = D and 0 = A. Thus, (1, 1) is ~~sustained reversal~~ an immediate switch maintained at the endpoint; (1, 0) immediate only; (0, 1) delayed; and (0, 0) no reversal. Parentheses identify the corresponding hypotheses; H2 concerns transfers and is not shown.

5 Randomization, Sample, and Power

The planned sample comprises 300 completed participants in ~~28–30~~ 20–60 matching pools, depending on attendance. ~~Ten-person pools are the default; no more than four fifteen-person pools are used, preserving at least 28 independent treatment assignments~~ Recruitment aims for larger cohorts, with a five-person session available when attendance is low. Treatment is assigned only after everyone finishes Block 1. ~~Within each session, pools are ordered by their final-round Block-1 A vote share and paired~~ Within each pool size, consecutive pools across sessions enter four-pool randomization sets. ~~One pool in each pair receives recovery and the other persistence, at random; an unmatched pool is assigned by a fair draw~~ Each set contains two recovery and two persistence assignments in random order. ~~Pairing improves balance in the~~

~~number eligible to reverse, which is measured before assignment~~ A persistent coded allocation record stores the set and position consumed after Block 1. Completed sets balance treatment within pool size; an incomplete final set retains its random prefix, so balance is not imposed within each session or guaranteed in the final sample. The maximum payment of EUR 15 implies a participant-compensation ceiling of EUR 4,500 for 300 completed participants. One round is selected from each block, so the number of rounds does not determine the payment ceiling.

Power depends on Block-1 endpoint support for A, the persistence-group reversal rate, the similarity of outcomes within pools, and the realised mix of pool sizes. ~~With 300 participants, the design yields 28–30 independent treatment assignments—approximately 14–15 pools per condition—because a larger pool supplies more individual outcomes but still receives only one treatment draw~~ The 300 participants provide 20–60 pool-level randomization units rather than 300, with approximately half in each condition. Five-person pools add units but may generate stronger dependence through shared group histories. Repeated rounds improve measurement and reveal trajectories; they do not create additional independent assignments.

At fixed eligibility, reversal rates, and within-pool dependence, endpoint power is unchanged by the number of follow-up rounds. Five Block-2 rounds instead limit how much post-assignment experience and within-session persistence can be observed; simulations of behavioral adjustment require explicit assumptions about its speed.

Pilot estimates will inform simulations spanning ~~28–30 pools~~ mixed pool sizes and stronger dependence in five-person pools and the observed number eligible to reverse. The primary randomization test reproduces the actual ~~paired and unmatched assignments~~ four-pool sets and incomplete prefixes within each pool size. Regression estimates cluster uncertainty by pool, adjust for ~~the prespecified pool-size indicator~~ prespecified pool-size indicators, and use small-sample inference; an equal-pool-weight analysis checks whether results depend on larger pools contributing more participants. ~~Each oTree session remains a separate interaction and randomization cohort~~ Interactions remain confined to each matching pool, while treatment balancing spans sessions. Where estimable, a sensitivity analysis excludes five-person pools without interpreting the fixed-versus-rematched comparison as causal.

Strategic decisions do not time out automatically. A visible 90-second countdown reminds participants to make a choice and click the confirmation button; at zero it remains a reminder and leaves the decision open. The software records response time and flags decisions exceeding 90 seconds, but never generates a ballot or allocation. The ~~laboratory manager~~ researcher or trained research assistant monitors the session and prompts delayed participants to complete the current decision so the group can advance.

6 Implementation Plan

The study will be conducted after ethical approval at the UTU Choice Lab decision-making laboratory ~~and may also be run at Tampere University's Decision-making Laboratory (DMLab) to accelerate recruitment~~. Adults register through ~~the host laboratory's participant system~~ UTU RPP/TOKA and receive its general rules and privacy information. ~~UTU Choice Lab uses UTU RPP/TOKA; DMLab uses ORSEE. At either site,~~ At UTU Choice Lab, oTree provides project-specific information and records a short affirmative study consent before practice. The ~~laboratory manager~~ researcher or trained research assistant then fixes a consenting roster of ~~10, 15, or 20~~ 5, 10, 15, or 20, forms the matching pools shown in Table 1, and handles any surplus arrivals under the ~~host laboratory's~~ study's procedure. No participant enters or changes pools after paid play begins.

The protocol will be ~~piloted before production~~ technically tested before data collection. A technical pilot will test ~~10- and 15-person~~ 5-, 10-, and 15-person pools, mixed-pool sessions, both Block-2 conditions, wait pages, ~~rematching~~ fixed-five interaction and rematching in larger pools, response-time flags, browser recovery, exports, and payment calculations. ~~A behavioral~~

~~pilot~~ The first complete ten-person session, included within the planned 300 participants, will assess comprehension, session length, choice distributions, fatigue, and expected earnings. It offers an optional six-question feedback form after everyone has finished the paid decisions and main questionnaires, before the debrief and payment display. The pretest aims to obtain five to ten voluntary responses about clarity, usability, suggested improvements, and the perceived study purpose. The form uses a separate common address and stores only aggregate ratings and independent shuffled text lists, without participant codes or links to experimental responses. It is disabled after this pretest; its questions appear in Attachment 7. The protocol, analysis plan, and data dictionary will ~~then be frozen before confirmatory data collection~~ be versioned before collection. Any substantive revision will undergo the required review before use, and the analysis plan will specify how pretest observations are handled.

Production sessions will last approximately 60 minutes. The ~~laboratory manager~~ researcher or trained research assistant will monitor synchronized progress and prompt delayed participants; the software will record delays but will never generate a ballot, proposal, approval decision, or allocation. A structured session log will record late arrivals, early departures, technical interruptions, staff interventions, and payment exceptions. The research team will review the log before applying preregistered data-quality rules.

7 Recruitment, Participant Care, and Resources

~~The host laboratory~~ UTU Choice Lab will invite adults through its participant system using a neutral description of a paid study of group decision making and public-service provision. Recruitment will not disclose the directional hypotheses. Participation is voluntary and will not affect employment, studies, or access to University services. The fixed participation payment is EUR 10; points earned in one randomly selected round from each block determine a performance bonus of up to EUR 5.

Foreseeable risks are minimal. Participants may experience mild discomfort with unequal allocations, the fictional crisis, or waiting for an interacting group. Instructions avoid moral or partisan labels, individual ballots remain private, and only group-level outcomes are shown to other participants. Before the task, participants are told that the detailed research question will be explained after paid decisions. This limited incomplete disclosure is necessary because identifying the hypothesis about weakening and restoring approval could influence the institutional ballot itself; it does not withhold task rules, payments, data processing, foreseeable risks, or the right to stop. The debrief explains the research interpretation immediately after paid decisions and emphasizes that no individual choice is treated as evidence of corruption, authoritarianism, or opposition to democracy.

Participant compensation is EUR 10–15, with a ceiling of EUR 4,500 for 300 completed participants. Laboratory charges and arrangements for timely arrivals who cannot enter a valid roster follow the agreement with ~~the host laboratory~~ UTU Choice Lab. Secure data services and routine dissemination costs follow available project funds. The experiment uses the open-source oTree platform and requires no proprietary experimental-software licence.

8 Data Governance, Schedule, and Outputs

The experimental interface uses participant codes rather than names. The production oTree study and coded research data are hosted on University of Turku IT servers in Finland~~, including for sessions conducted at DMLab~~. ~~Each host laboratory separately administers the identifying information needed for recruitment, attendance, and payment under its own privacy notice. The research team receives only coded experimental data and never receives any temporary identity-to-code link; the host laboratory removes that link after payment under its procedure.~~ UTU RPP/TOKA separately administers recruitment and attendance under its own privacy

notice. It does not assign oTree codes, receive experimental responses, or administer payment. The research team makes immediate payment with a separate code-and-amount list; any named payment record contains no oTree code. The list is destroyed after payment reconciliation and no later than 30 November 2026. Anonymisation also requires removing or coarsening identifying combinations and a documented disclosure-risk review. Until that review confirms anonymisation, coded data remain restricted personal data. Access to coded research data is limited to Janne Tukiainen, Hector Bahamonde, and authorised personnel. Reporting will use aggregate or de-identified data, while ~~coded~~ anonymised data may be retained for verification and compatible future research under the research privacy notice.

Table 3: Planned project schedule

Period	Main activities
September 2026	Ethical review, final programming and translation, ~~technical and behavioral pilots~~ technical testing, data-protection review, and preregistration.
1–31 October 2026	Recruitment and laboratory data collection at UTU Choice Lab, including the first ten-person behavioral pretest.
November–December 2026	Payment reconciliation and deletion of the temporary payment list by 30 November, followed by disclosure-risk review and data verification, de-identification, reproducible cleaning, primary and secondary analyses, and robustness checks.
January–March 2027	Manuscript preparation and preparation of de-identified replication materials.
After reporting	Retention, controlled archiving, or deletion in accordance with the approved data management plan and privacy notice.

The main outputs will be academic articles, presentations, and replication materials. Consistent with University open-science policy, code, documentation, and data that can be shared safely will be made as open as possible and as restricted as necessary. The primary conclusion will follow the estimand and decision rule stated above; related econometric and randomization-based methods may be used to assess precision and robustness without changing the randomized comparison.

References

Bermeo, Nancy (2016). "On Democratic Backsliding." In: *Journal of Democracy* 27.1, pp. 5–19.

Claassen, Christopher (2020). "In the Mood for Democracy? Democratic Support as Thermostatic Opinion." In: *American Political Science Review* 114.1, pp. 36–53.

Dal Bó, Pedro, Andrew Foster, and Louis Putterman (2010). "Institutions and Behavior: Experimental Evidence on the Effects of Democracy." In: *American Economic Review* 100.5, pp. 2205–2229.

Dohmen, Thomas et al. (2011). "Individual Risk Attitudes: Measurement, Determinants, and Behavioral Consequences." In: *Journal of the European Economic Association* 9.3, pp. 522–550.

Fehr, Ernst and Simon Gächter (2000). "Cooperation and Punishment in Public Goods Experiments." In: *American Economic Review* 90.4, pp. 980–994.

Fischbacher, Urs and Simon Gächter (2010). "Social Preferences, Beliefs, and the Dynamics of Free Riding in Public Goods Experiments." In: *American Economic Review* 100.1, pp. 541–556.

Forteza, Alvaro, Irene Mussio, and Juan S. Pereyra (2024). "Can Political Gridlock Undermine Checks and Balances? A Lab Experiment." In: *Journal of Behavioral and Experimental Economics* 108, p. 102128.

Glaeser, Edward L. et al. (2000). "Measuring Trust." In: *Quarterly Journal of Economics* 115.3, pp. 811–846.

Goodman, Sara Wallace (2021). "Immigration Threat, Partisanship, and Democratic Citizenship: Evidence from the US, UK, and Germany." In: *Comparative Political Studies* 54.11, pp. 2052–2083.

Graham, Matthew and Milan Svolik (2020). "Democracy in America? Partisanship, Polarization, and the Robustness of Support for Democracy in the United States." In: *American Political Science Review* 114.2, pp. 392–409.

Hamman, John R., Roberto A. Weber, and Jonathan Woon (2011). "An Experimental Investigation of Electoral Delegation and the Provision of Public Goods." In: *American Journal of Political Science* 55.4, pp. 738–752.

Hetherington, Marc J. and Elizabeth Suhay (2011). "Authoritarianism, Threat, and Americans' Support for the War on Terror." In: *American Journal of Political Science* 55.3, pp. 546–560.

Markussen, Thomas, Louis Putterman, and Jean-Robert Tyran (2014). "Self-Organization for Collective Action: An Experimental Study of Voting on Sanction Regimes." In: *Review of Economic Studies* 81.1, pp. 301–324.

Saikkonen, Inga A.-L. and Henrik Serup Christensen (2023). "Guardians of Democracy or Passive Bystanders? A Conjoint Experiment on Elite Transgressions of Democratic Norms." In: *Political Research Quarterly* 76.1, pp. 127–142.

Sutter, Matthias, Stefan Haigner, and Martin G. Kocher (2010). "Choosing the Carrot or the Stick? Endogenous Institutional Choice in Social Dilemma Situations." In: *Review of Economic Studies* 77.4, pp. 1540–1566.